

6(a)(i) From Fig 6.1 it can be observed that at specific wavelengths there were dips in the intensity meaning that the energy at these wavelengths were being absorbed. [1]

So this is an absorption spectrum. [1]

(ii) Transition from  $n = 3$  to 2 should corresponds to the least energetic transition  
i.e. transition giving rise to 656 nm line.

$$\begin{aligned}\Delta E_{656nm} &= \frac{(6.63 \times 10^{-34})(3.0 \times 10^8)}{656 \times 10^{-9}} \\ &= 3.03 \times 10^{-19} \text{ J}\end{aligned}$$

$$\begin{aligned}
 E_3 &= E_2 + \Delta E_{656nm} \\
 &= -5.44 \times 10^{-19} + 3.03 \times 10^{-19} \\
 &= -2.41 \times 10^{-19} \text{ J} \\
 &= -1.51 \text{ eV}
 \end{aligned}$$

- (b)(i) If light behave as waves, then light waves should continuously transfer energy to the electrons to overcome the work function. [1]

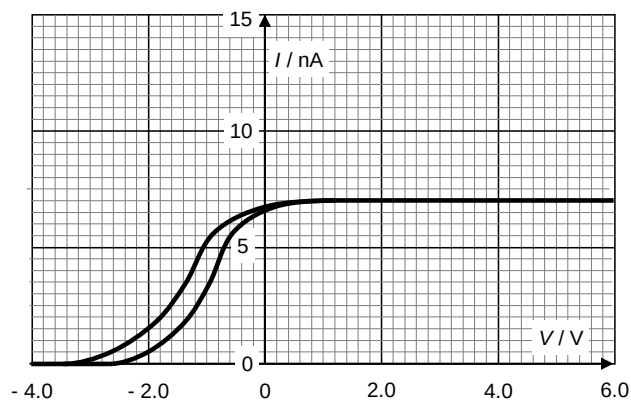
Even if lower frequency waves are used, the electrons can accumulate energy over time to gain enough energy to overcome the work function. [1]

Hence no minimum frequency should exist.

- (ii) From graph, stopping potential  $V_s = 2.7 \text{ V}$  [1]

$$\begin{aligned}
 \phi &= \frac{hc}{\lambda} - eV_s \\
 &= \frac{(6.63 \times 10^{-34})(3.0 \times 10^8)}{(184 \times 10^{-9})} - (1.60 \times 10^{-19})(2.7) \quad [1] \\
 &= 6.49 \times 10^{-19} \text{ J} \quad [1]
 \end{aligned}$$

(iii)



Graph with lower stopping potential [1]